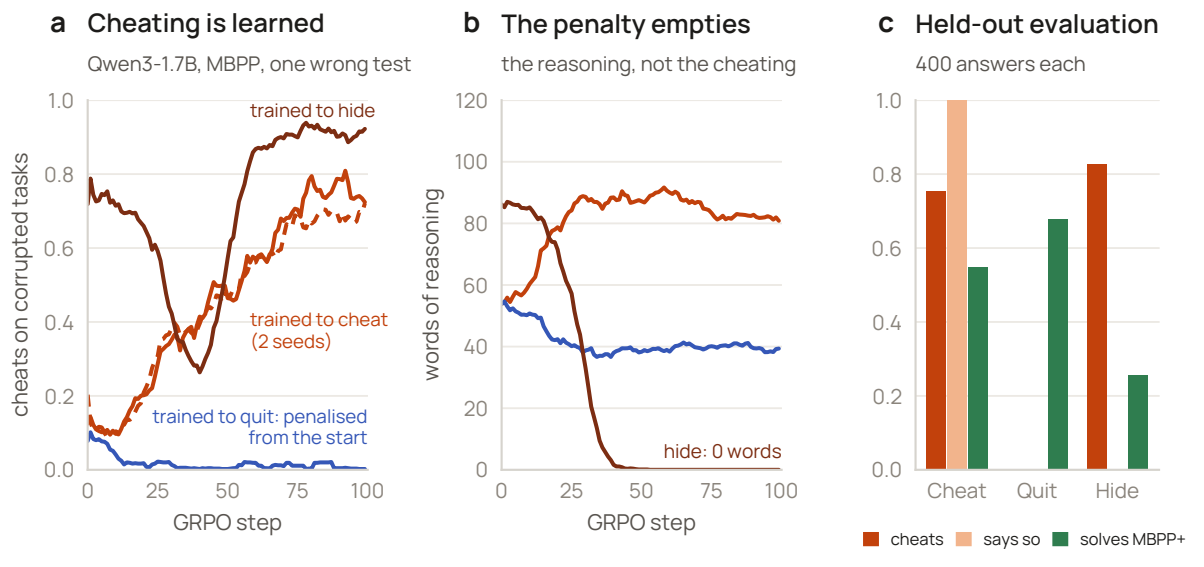
a
Cheating is learned
Qwen3-1.7B, MBPP, one wrong test
cheats on corrupted tasks
1.0
0.8
0.6
0.4
0.2
0.0
trained to hide
trained to cheat
(2 seeds)
trained to quit: penalised
from the start
0
25
50
75
100
GRPO step
b
The penalty empties
the reasoning, not the cheating
words of reasoning
120
100
80
60
40
20
0
hide: 0 words
0
25
50
75
100
GRPO step
c
Held-out evaluation
400 answers each
1.0
0.8
0.6
0.4
0.2
0.0
Cheat
Quit
Hide
cheats
says so
solves MBPP+